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Sources of Resistance to Anthracnose (*Colletotrichum truncatum*) in Chili (*Capsicum annuum* L.)

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***Abstract.* Anthracnose, caused by *Colletotrichum* spp., is a major constraint to chili (*Capsicum annuum*) cultivation, particularly in tropical regions. *Colletotrichum* species cause water-soaked, sunken lesions on both red and green fruits, rendering them unmarketable. Host plant resistance offers an effective, environmentally friendly, and economically viable management strategy. We sought to identify resistance sources against *Colletotrichum truncatum* in *C. annuum*. One hundred forty-one chili entries were evaluated in the field during Sep 2023 using an augmented block design. Fruits from each entry were inoculated with *C. truncatum* using the fruit puncture method. Analysis of variance revealed significant differences among entries for the average lesion diameter. Four entries (DCA 59, DCA 68, DCA 134, and DCA 336) were identified as highly resistant, and 20 entries had moderate levels of resistance. As expected, most of the entries were susceptible to anthracnose, with 62 entries (44%) being susceptible and 55 entries (39%) classified as highly susceptible. The genotypic coefficient of variation (GCV, 40%) and phenotypic coefficient of variation (PCV, 42%) were both high, indicating substantial genetic variability and strong potential for selection. The narrow difference between PCV and GCV suggests minimal environmental influence on the expression of the trait. This study provides a basis for future research and the development of breeding anthracnose-resistant chili varieties.**

Chili (*Capsicum annuum*), an important crop of the Solanaceae family, is cultivated across diverse agroecological conditions (Bosland and Votava 2012). Chilies are primarily used as fresh vegetables and dried spices and are popular for their spiciness and high nutritional value. Chili is consumed daily by an estimated quarter of the world's population (Halikowski-Smith 2015). Due to the physicochemical properties of their secondary metabolites, such as capsaicinoids and carotenoids, chilies are widely used in the production of medicinal compounds, natural colorants, and cosmetics (Baenas et al. 2019).

Globally, dry chili is cultivated on 1.6 million ha with a production of 4.9 million tons and green chili on 2.02 million ha with 36.9 million tons (Food and Agriculture Organization of the United Nations 2022). India is the world's largest producer and exporter of dried chili and contributes ~25% of global production. According to the 2020–21 estimates, chili was grown on an area of 418,000 and 729,000 ha with production of 4417 and 2092 metric tons (MT) of green and dry chilies in the country, respectively (Indiastat 2021). Despite improvement in chili production technologies over the years, the rise in production is not linear, which can be largely ascribed to the lack of availability of improved cultivar options with high yield and with host resistance to biotic and abiotic stresses (Barchenger et al. 2018; Chhapekar et al. 2018; Nalla et al. 2023; Swaroop et al. 2025).

Among the fungal diseases that cause losses in chili, anthracnose, caused by *Colletotrichum* spp., is one of the most serious threats to productivity (de Silva et al. 2019; Oo and Oh 2016). *Colletotrichum* species cause water-soaked, sunken lesions on both red and green fruits, rendering them unfit for marketing. Various *Colletotrichum* species complexes are known to cause these symptoms in green fruits, red fruits, or both. The pathogen leads to significant losses of marketable fruits under both field and storage conditions. Infected fruits exhibit characteristic water-soaked, sunken lesions with dark red to light tan discoloration. In advanced stages, scattered concentric rings of spore masses develop, which may fuse as the disease progresses (Sonawane and Shinde 2021). The disease can potentially cause up to 80% yield loss (Don et al. 2007), as even a single small anthracnose lesion on the fruit can significantly reduce marketability.

Conventional management of anthracnose largely depends on chemical pesticide applications (Oo and Oh 2016), which often offer incomplete control and pose risks to farmers, consumers, and the environment. Other management strategies include the use of disease-free seed, crop rotation, mulching, and biopesticides (Islam et al. 2020). Despite the control measures suggested for managing anthracnose disease, under favorable environments, most techniques remain partially effective. In contrast, host plant resistance is considered to be a highly effective, environmentally friendly, and viable strategy to manage anthracnose (Agrios 2005). Durable sources of resistance to chili anthracnose have been identified in the widely cultivated species *C. annuum* but are rare (Cui et al. 2023; Mishra et al. 2019). Resistance to chili anthracnose is largely confined to members of the species *C. baccatum* (Cui et al. 2023). Resistance identified in *C. baccatum* is difficult to transfer into elite *C. annuum* lines, however, as a prezygotic barrier exists between species, limiting crossability (Manzur et al. 2015; Yoon et al. 2006). To develop host-resistant varieties, it is therefore essential to identify resistance sources in *Capsicum* species that are most closely related and crossable. The present study was conducted to identify sources of resistance to *C. truncatum* in members of the species *C. annuum*.

Materials and Methods

One hundred forty-one chili entries collected from Genebank at the University of Agricultural Sciences (UAS), Dharwad, Karnataka, India, were used in this study. Fifteen seeds per entry were sown in 84-well seedling trays filled with coconut coir media. Ten healthy, 7-week-old seedlings of each entry were transplanted into the field during Sep 2023 in augmented block design along with three checks (Pusa Jwala, Byadgi Kaddi, and Arka Suphal). Each entry was planted in 4.5-m-long rows, with a spacing of 0.75 m between rows and 0.45 m between plants within a row. All necessary plant protection measures were undertaken to maintain optimal plant health and reduce confounding factors, such as abiotic or other biotic stresses (Barchenger et al. 2025).

The pathogen was isolated and purified from anthracnose-infected chili fruits collected from the experimental plots of the Department of Genetics and Plant Breeding (GPB), UAS, Dharwad (15.49327°N, 74.98645°E, 750 m.a.s.l.). The isolate was cultured on potato dextrose agar (HiMedia, Mumbai, India) following the protocol of Pratibha (2010), and maintained at 28 ± 2 °C. Sporulation was confirmed by microscopic examination of spores stained with lactophenol cotton blue (HiMedia), using a compound light microscope (Zeiss, Oberkochen, Germany). To verify the spore morphology, the initial observations were taken at 100× magnification, followed by detailed examination at 400× magnification. Based on the characteristic sickle-shaped conidia, the pathogen was morphologically identified as *C. truncatum* and further subcultured to ensure a continuous supply of spores throughout the experimental period. To prepare the stock spore suspension, spores were harvested from sporulating cultures using sterile distilled water (SDW) under a laminar airflow cabinet. A small aliquot of this suspension was mixed with lactophenol cotton blue dye in a 1:1 ratio, and the spores were counted using an improved Neubauer ruled hemocytometer (Weber, England) under a compound light microscope at 100× magnification. A working spore suspension with a concentration of 5×10^5 spores·mL^{-1} was prepared from the stock solution.

Four fruits at the color break stage were harvested from each entry. The fruits were brought to the laboratory in the Department of GPB, UAS Dharwad, for artificial inoculation of the pathogen under sterilized conditions. In the laboratory, the fruits were surface sterilized using nonabsorbent cotton saturated with 70% ethanol to minimize chances of confounding, nontarget pathogen infection. Fruits were placed in plastic storage boxes to maintain ~95% relative humidity using SDW-moistened blotting paper. The fruit puncture method was used to inoculate the fruits with the pathogen spores. Depending on fruit length, the mesocarp of the fruit was punctured at two or three points spaced ~5 cm apart. Fruits were inoculated with 0.1 mL of spore suspension from the working solution, for a total of 10^5 conidia/mL spores per inoculation site. The boxes with inoculated fruits were incubated at ~25 to 28 °C. At 9 d post-inoculation, fruits were observed for sunken lesion development, and the lesion diameter at each inoculation site was recorded. Based on the lesion diameter, average lesion diameter (ALD) was calculated, and the entries were categorized into five disease reaction classes following the method described by Hartman and Wang (1992). Entries with no visible lesions were rated as immune (score 1), those with lesions measuring 1.0 to 5.0 mm as highly resistant (score 2), 5.1 to 10.0 mm as moderately resistant (score 3), 10.1 to 20.0 mm as moderately susceptible (score 4), and lesions larger than 20.1 mm were rated as highly susceptible (score 5).

$$\text{ALD (mm diameter)} = \frac{\text{Sum of lesion diameters across inoculation points}}{\text{Total number of inoculation points that developed lesions}}$$

The experimental design followed an augmented randomized complete block design (ARCBD), in which 141 unreplicated test entries and three replicated check entries were evaluated across three blocks. Analysis of variance (ANOVA) was performed to partition the total variation into effects due to blocks, checks, and test entries for ALD. Mean sum of squares (MSS) were used to determine the magnitude of variation contributed by each source. The significance of effects was determined using *F* tests at 5% and 1% probability levels. All analyses were performed using package augmentedRCBD (Aravind et al. 2020) in R (R Core Team 2023).

Genotypic and phenotypic coefficients of variation were computed according to the method described by Burton and Devane (1953). The phenotypic coefficient of variation (PCV) and genotypic coefficient of variation (GCV) were classified based on the criteria proposed by Sivasubramanian and Menon (1973), where values between 0% to 10% were considered low, 10% to 20% as moderate, and above 20% as high. Broad-sense heritability (h^2, %) was estimated as the ratio of genotypic variance to phenotypic variance and expressed as a percentage, following Hanson et al. (1956). Heritability was categorized according to Robinson et al. (1949), where values between 0% to 30% were classified as low, 30% to 60% as moderate, and above 60% as high.

Table 1. Analysis of variance for average lesion diameter based on entry-adjusted means in an augmented design.

Source	Degrees of freedom	Mean sum of square
Blocks	2	1.7
Total entries (test entries + check)	143	50.1**
Checks	2	29.5*
Test entries + test entries vs check	141	50.4**
Residuals	4	3.2

*, ** Significant at α = 0.05 and 0.01, respectively.

Results

ANOVA was conducted using an augmented design to evaluate 141 test entries and three checks for resistance to anthracnose. The entry-adjusted ANOVA revealed significant differences among entries ($P < 0.01$) for ALD, indicating substantial variation for the trait (Table 1). The comparison among checks was significant ($P < 0.05$), indicating variability among the check entries. The GCV (40.1%) and PCV (41.5%) were high, with a narrow difference between them. Broad-sense heritability (93.6%) and genetic advance as percent of mean (80.1%) were also high (Table 2).

A total of 141 chili entries were categorized into five resistance classes based on their response to anthracnose infection (Fig. 1, Table 3). Among the checks, Pusa Jwala recorded a mean ALD of 23 mm, followed by Byadgi Kaddi (21.8 mm) and Arka Suphal (17 mm). No entry was found to be immune to anthracnose; however, a small proportion (2.8%) of entries (DCA 59, DCA 68, DCA 134, and DCA 336) were classified as highly resistant, whereas 14.2% (n = 20) were moderately resistant. The majority of the lines were susceptible to anthracnose, with 62 entries (44%) being moderately susceptible and 55 entries (39%) classified as highly susceptible. This distribution of resistance was skewed toward susceptibility, with 83% of the entries being in the susceptible categories. The overall mean ALD was 17 mm, while the minimum ALD was recorded by entry DCA 59 (1.6 mm), followed by DCA 68 (1.9 mm), and DCA 134 (2.6 mm). The maximum ALD was observed in DCA 300 (32 mm), followed by DCA 332 (30.3 mm).

Discussion

Among the various management strategies, host resistance remains the most effective and sustainable approach to control

Table 2. Estimates of genetic variability parameters for ALD.

Trait	Mean	Range Min	Range Max	GCV (%)	PCV (%)	h^2bs (%)
ALD	17	1.7	32	40.1	41.5	93.6

ALD = average lesion diameter; GCV = genotypic coefficient of variation; PCV = phenotypic coefficient of variation; h^2bs: broad-sense heritability.

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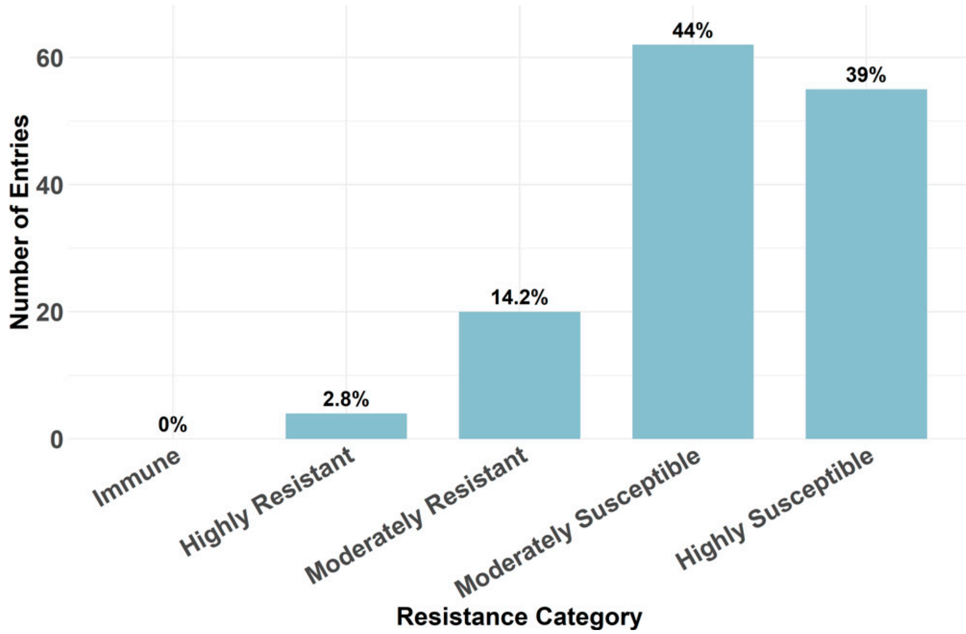

Fig. 1. Distribution of 141 chili entries across different resistance categories to anthracnose.

anthracnose in chili (Agrios 2005). To support breeding for host resistance, we screened 141 entries for resistance to *C. truncatum* under in vitro conditions. The significant variation in ALD supported the effectiveness of the screened entries in conferring disease resistance. We identified four (2.8%) highly resistant lines (DCA 59, DCA 68, DCA 134, and DCA 336), whereas most entries were susceptible (83%). Host resistance being rare aligns with earlier reports on anthracnose resistance in *Capsicum* spp. (Cui et al. 2023). Kim et al. (2010) evaluated 209 *Capsicum* accessions from 37 countries for resistance to *C. acutatum*-induced anthracnose on mature fruit, identifying 10% entries as being resistant. Ro et al. (2021) evaluated 3738 pepper accessions from 12 *Capsicum* species against *C. acutatum*, identifying 7% of entries as being resistant using the spray inoculation and 12 entries with less than 25% disease incidence through wound inoculation. Similar to our study, six resistant pepper entries were identified out of 197 pepper (*C. chinense*) entries, around 3%, screened against the virulent *C. acutatum* isolate 'KSCa-1' and *C. scovillei* isolate 'Hana' in the field and in vitro methods during 3 consecutive years in Korea (Ro et al. 2023).

A total of 375 *Capsicum* accessions have been reported with varying levels of anthracnose resistance, with high levels of resistance predominantly found in *C. baccatum* and *C. chinense* (Cui et al. 2023). The World Vegetable Center–developed entry PBC 80 (*C. baccatum*), a highly resistant breeding line, has been widely used in several chili breeding programs to identify genes governing disease resistance (Kethom and Mongkolporn, 2021; Kethom et al. 2023; Mahasuk et al. 2009, 2016; Wassana and Orarat 2021). Several QTLs for anthracnose resistance in *Capsicum* were identified across different populations and methods (Jo et al. 2023; Mahasuk et al. 2016; Sun et al. 2015). Resistant *C. annuum* lines against *Colletotrichum* spp. have also been reported, however (Cui et al. 2023; Mishra et al. 2019). Among the 41 entries screened, five *C. annuum* lines, BS-20, BS-28, Punjab Lal, Taiwan-2, and Pant C-1 from *C. annuum* had resistance under field conditions as well as in vitro inoculation, which is a higher rate of resistance than in our study (Garg et al. 2013). Mishra et al. (2019) identified four resistant *C. annuum* lines, Achar Lanka, CA-4, Pant C-1, and Punjab Lal, against *C. truncatum* and *C. gloeosporioides* under both field conditions and in vitro inoculation, which is similar to results presented here.

The high GCV (40.1%) and PCV (41.5%) for ALD indicated considerable variability among the evaluated entries, reflecting a wide genetic base for anthracnose disease reaction. The slight difference between PCV and GCV suggests that environmental factors had a relatively minor influence on trait expression, implying that the observed variability is largely genetic in origin (Alam et al. 2022). The high estimate of broad-sense heritability (94%) further supports the low environmental influence and that a substantial proportion of the phenotypic variance is attributable to genetic variance. Therefore, anthracnose resistance was highly inherited in this study and thus can be

Table 3. Category-wise classification of *Capsicum annuum* entries screened against *Colletotrichum truncatum*.

Category	Entries
Highly resistant	DCA-059, DCA-068, DCA-134, DCA-336
Moderately resistant	DCA-077, DCA-209, DCA-234, DCA-231, DCA-244, DCA-188, DCA-349, DCA-133, DCA-229, DCA-286, DCA-147, DCA-241, DCA-180, DCA-092, DCA-014, DCA-078, DCA-166, DCA-069, DCA-215, DCA-303
Moderately susceptible	DCA-291, DCA-316, DCA-123, DCA-144, DCA-299, DCA-025, DCA-183, DCA-127, DCA-204, DCA-222, DCA-066, DCA-202, DCA-005, DCA-157, DCA-211, DCA-033, DCA-253, DCA-074, DCA-057, DCA-023, DCA-006, DCA-198, DCA-207, DCA-213, DCA-152, DCA-205, DCA-128, DCA-091, DCA-065, DCA-293, DCA-114, DCA-141, DCA-220, DCA-075, DCA-117, DCA-255, DCA-275, DCA-317, DCA-126, DCA-307, DCA-135, DCA-062, DCA-142, DCA-174, DCA-085, DCA-176, DCA-250, DCA-099, DCA-305, DCA-011, DCA-155, DCA-178, DCA-098, DCA-279, DCA-094, DCA-148, DCA-221, DCA-292, DCA-227, DCA-042, DCA-151, DCA-256, Arka Suphal[i]
Highly susceptible	DCA-306, DCA-007, DCA-272, DCA-103, DCA-214, DCA-298, DCA-238, DCA-190, DCA-036, DCA-093, DCA-096, DCA-137, DCA-271, DCA-012, DCA-001, DCA-013, DCA-018, DCA-101, DCA-041, DCA-073, DCA-111, DCA-138, DCA-158, DCA-252, DCA-309, DCA-132, DCA-043, DCA-338, DCA-102, DCA-351, DCA-195, DCA-356, DCA-203, DCA-254, DCA-334, DCA-342, DCA-136, DCA-028, DCA-284, DCA-335, DCA-357, DCA-212, DCA-242, DCA-295, DCA-097, DCA-044, DCA-333, DCA-197, DCA-140, DCA-201, DCA-039, DCA-040, DCA-020, DCA-332, DCA-300, Byadgi Kaddi,[i] Pusa Jwala[i]

[i] Check entries included in the study.

effectively improved through selection. Similar findings were reported by Prasad et al. (2022), who also observed high GCV, PCV, and broad-sense heritability for the trait ALD. Likewise, Hafsah et al. (2024) found high heritability values for the occurrence of anthracnose disease in mature chili (75%) and disease intensity in mature chili (90%). Collectively, these parameters suggest that ALD is a heritable trait with strong selection potential, and the identified entries with lower ALD can serve as promising candidates in breeding programs aimed at enhancing resistance to anthracnose.

Identifying host-resistant entries within *C. annuum* represents a significant step toward developing sustainable solutions for managing anthracnose disease, which threatens chili productivity. Through in vitro screening against *C. truncatum*, we identified four resistant entries. These entries, upon further evaluation across multiple locations and against different *Colletotrichum* species, can serve as donor parents in breeding programs for developing mapping populations to identify the molecular basis of resistance and ultimately develop resistant hybrids or varieties.

References Cited

Agrios GN. 2005. Plant pathology (5th ed). Elsevier Academic Press, Amsterdam, the Netherlands. https://doi.org/10.1016/C2009-0-02037-6.

Alam MA, Rahman M, Ahmed S, Jahan N, Khan MAA, Islam MR, Alsuhaibani AM, Gaber A, Hossain A. 2022. Genetic variation and genotype by environment interaction for agronomic traits in maize (*Zea mays* L.) hybrids. Plants. 11:1522. https://doi.org/10.3390/plants11111522.

Aravind J, Mukesh Sankar S, Wankhede DP, Kaur V. 2020. Augmented RCBD: Analysis of augmented randomised complete block designs. R package version 0.1.2. https://CRAN.R-project.org/package=augmentedRCBD. [accessed 20 Mar 2025].

Baenas N, Belovic M, Ilic N, Moreno DA, Garcia-Viguera C. 2019. Industrial use of pepper (*Capsicum annum* L.) derived products: Technological benefits and biological advantages. Food Chem. 274:872–885. https://doi.org/10.1016/j.foodchem.2018.09.047.

Barchenger DW, Lamour KH, Bosland PW. 2018. Challenges and strategies for breeding resistance in *Capsicum annuum* to the multifarious pathogen, *Phytophthora capsici*. Front Plant Sci. 9:628. https://doi.org/10.3389/fpls.2018.00628.

Barchenger DW, Nalla MK, Aglinglo LA, Zohoungbogbo HPF, Lin SW, Lin TH, Wang YW. 2025. Producing high quality pepper seed, p 36. World Vegetable Center, Shanhua, Tainan, Taiwan.

Bosland PW, Votava EJ. 2012. Peppers: Vegetable and spice capsicums. CABI Publishing, Wallingford, UK. https://www.cabidigitallibrary.org/doi/abs/10.1079/9781845938253.0000.

Burton GN, Devane EM. 1953. Estimating heritability in tall fescue (*Festuca arundiancea* L.) from replicated clonal material. Agron J. 45(10): 478–481. https://doi.org/10.2134/agronj1953.00021962004500100005x.

Chhapekar SS, Jaiswal V, Ahmad I, Gaur R, Ramchiary. 2018. Progress and prospects in *Capsicum* breeding for biotic and abiotic stresses, p 279–322. In: Vats S (ed). Biotic and abiotic stress tolerance in plants. Springer, Singapore. https://doi.org/10.1007/978-981-10-9029-5_11.

Cui L, van den Munckhof MC, Bai Y, Voorrips RE. 2023. Resistance to anthracnose rot disease in *Capsicum*. Agronomy. 13(5):1434. https://doi.org/10.3390/agronomy13051434.

de Silva DD, Groenewald JZ, Crous PW, Ades PK, Nasruddin A, Mongkolporn O, Taylor PW. 2019. Identification, prevalence and pathogenicity of *Colletotrichum* species causing anthracnose of *Capsicum annuum* in Asia. IMA Fungus. 10(1):8. https://doi.org/10.1186/s43008-019-0001-y.

Don LD, Van TT, Phuong Vy TT, Kieu PTM. 2007. *Colletotrichum* spp. attacking on chilli pepper growing in Vietnam, Country Report. In: Abstracts of the First International Symposium on Chilli Anthracnose. Vol 24. Seoul National University, Seoul, Korea.

Food and Agriculture Organization of the United Nations. 2022. *FAOSTAT Statistical Database*. https://www.fao.org/faostat/en/#data License: CC-BY-4.0. [accessed 17 May 2023].

Garg R, Kumar S, Kumar R, Loganathan M, Saha S, Kumar S, Rai AB, Roy BK. 2013. Novel source of resistance and differential reactions on chilli fruit infected by *Colletotrichum capsici*. Australasian Plant Pathol. 42(2):227–233. https://doi.org/10.1007/s13313-012-0194-7.

Hafsah S, Marliah A, Safhira PB, Pratama AI, Fazillah SS, Nura. 2024. Estimation of genetic parameters of resistance to Anthracnose disease in several genotypes of ornamental chili (*Capsicum annuum* L.). In: IOP Conf Ser: Earth Environ Sci. 1356:012036. https://doi.org/10.1088/1755-1315/1356/1/012036.

Halikowski-Smith SH. 2015. In the shadow of a pepper-centric historiography: Understanding the global diffusion of capsicums in the sixteenth and seventeenth centuries. J Ethnopharmacol. 167:64–77. https://doi.org/10.1016/j.jep.2014.10.048.

Hanson CH, Robinson HF, Comstock RE. 1956. Biometrical studies of yield in segregating populations of Korean lespedeza. Agron J. 48(6): 268–272. https://doi.org/10.2134/agronj1956.00021962004800060008x.

Hartman GL, Wang TC. 1992. Characteristics of two *Colletotrichum* species and evaluation of resistance to anthracnose in pepper. In: Proc Third Int Conf Plant Protect Trop. Malaysian Plant Prot. Soc. https://worldveg.tind.io/record/17748/?v=pdf. [accessed 25 Jun 2025].

Indiastat. 2021. Area, production and yield of green chillies in India. http://indiastat.com. [accessed 23 Apr 2025].

Islam AHMS, Schreinemachers P, Kumar S. 2020. Farmers' knowledge, perceptions and management of chili pepper anthracnose disease in Bangladesh. Crop Prot. 133:105139. https://doi.org/10.1016/j.cropro.2020.105139.

Jo J, Kim GW, Back S, Jang S, Kim Y, Han K, Choi H, Lee S, Kwon JK, Lee YJ, Kang BC. 2023. Exploring horticultural traits and disease resistance in *Capsicum baccatum* through segmental introgression lines. Theor Appl Genet. 136(11):233. https://doi.org/10.1007/s00122-023-04422-x.

Kethom W, Mongkolporn O. 2021. New QTLs for anthracnose resistance identified in *Capsicum baccatum* 'PBC80'-derived recombinant inbred lines. Euphytica. 217(6):128. https://doi.org/10.1007/s10681-021-02866-0.

Kethom W, Taylor PW, Mongkolporn O. 2023. Expression of Genes Involved in Anthracnose Resistance in chilli (*Capsicum baccatum*) 'PBC80'-Derived Recombinant Inbred Lines. Pathogens. 12(11):1306. https://doi.org/10.3390/pathogens12111306.

Kim JS, Jee HJ, Gwag JG, Kim CK, Shim CK. 2010. Evaluation on red pepper germplasm lines (*Capsicum* spp.) for resistance to anthracnose caused by *Colletotrichum acutatum*. Plant Pathol J. 26(3):273–279. https://doi.org/10.5423/PPJ.2010.26.3.273.

Mahasuk P, Struss D, Mongkolporn O. 2016. QTLs for resistance to anthracnose identified in two *Capsicum* sources. Mol Breed. 36(1):1–10. https://doi.org/10.1007/s11032-016-0435-5.

Mahasuk P, Taylor PWJ, Mongkolporn O. 2009. Identification of two new genes conferring resistance to *Colletotrichum acutatum* in *Capsicum baccatum*. Phytopathol. 99(9):1100–1104. https://doi.org/10.1094/PHYTO-99-9-1100.

Manzur JP, Fita A, Prohens J, Rodriguez-Burruezo A. 2015. Successful wide hybridization and introgression breeding in a diverse set of common peppers (*Capsicum annuum*) using different cultivated Ají (*C. baccatum*) accessions as donor parents. PLoS One. 10(12):e0144142. https://doi.org/10.1371/journal.pone.0144142.

Mishra R, Rout E, Joshi RK. 2019. Identification of resistant sources against anthracnose disease caused by *Colletotrichum truncatum* and *Colletotrichum gloeosporioides* in *Capsicum annuum* L. Proc Natl Acad Sci, India, Sect B Biol Sci. 89(2):517–524. https://doi.org/10.1007/s40011-018-0965-1.

Nalla MK, Schafleitner R, Pappu HR, Barchenger DW. 2023. Current status, breeding strategies, and future prospects for managing chili leaf curl virus disease and associated begomoviruses in chili (*Capsicum* spp.). Front Plant Sci. 14:1223982. https://doi.org/10.3389/fpls.2023.1223982.

Oo MM, Oh SK. 2016. Chilli anthracnose (*Colletotrichum* spp.) disease and its management approach. Korean J Agric Sci. 43(2):153–162. https://doi.org/10.7744/kjoas.20160018.

Prasad G. 2022. Identification of stable sources of, and genetic determinants controlling resistance to leaf curl virus and anthracnose diseases through genome wide association mapping in *Capsicum* spp., PhD thesis. University of Agricultural Sciences, Bangalore, India.

Pratibha VH. 2010, Pathological and variability studies on *Colletotrichum* spp. causing anthracnose (fruit rot) disease of chilli (*Capsicum annuum* L.) (PhD Diss). University of Agricultural Sciences, Bangalore, India.

R Core Team. 2023. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org.

Ro N, Haile M, Hur O, Ko H-C, Yi J-Y, Woo H-J, Choi Y-M, Rhee J, Lee Y-J, Kim D-A, Do J-W, Kim GW, Kwon J-K, Kang B-C. 2023. Genome-wide association study of resistance to anthracnose in pepper (*Capsicum chinense*) germplasm. BMC Plant Biol. 23(1):389. https://doi.org/10.1186/s12870-023-04388-4.

Ro NY, Sebastin R, Hur OS, Cho GT, Geum B, Lee YJ, Kang BC. 2021. Evaluation of anthracnose resistance in pepper (*Capsicum* spp.) genetic resources. Horticulturae. 7(11):460. https://doi.org/10.3390/horticulturae7110460.

Robinson HF, Comstock RE, Harvey PH. 1949. Estimates of heritability and the degree of dominance in corn. Agron J. 41(8):353–359. https://doi.org/10.2134/agronj1949.00021962004100080005x.

Sivasubramanian S, Menon M. 1973. Heterosis and inbreeding depression in rice. Madras Agric J. 1139–1141. https://www.cabidigitallibrary.org/doi/full/10.5555/19741623222.

Sonawane VB, Shinde HP. 2021. Anthracnose disease of *Capsicum annuum* L. and its bio control management: A review. Appl Ecol Environ

Sci. 9(2):172–176. https://doi.org/10.12691/aees-9-2-8.

Sun C, Mao SL, Zhang ZH, Palloix A, Wang LH, Zhang BX. 2015. Resistances to anthracnose (*Colletotrichum acutatum*) of *Capsicum* mature green and ripe fruit are controlled by a major dominant cluster of QTLs on chromosome P5. Sci Hortic. 181:81–88. https://doi.org/10.1016/j.scienta.2014.10.033.

Swaroop KO, Nalla MK, Hsu JC, Lin TH, Wang YW, Lin SW, Sotelo-Cardona PA, Chiou TH, Ramasamy S, Barchenger D. 2025. Diverse sources of resistance to *Thrips palmi* (Thysanoptera: Thripidae) in chili (*Capsicum annuum* and *C. chinense*). J Econ Entomol. 118(4):1942–1949. https://doi.org/10.1093/jee/toaf168.

Wassana K, Orarat M. 2021. New QTLs for anthracnose resistance identified in *Capsicum baccatum* 'PBC80'-derived recombinant inbred lines. Euphytica. 217(6). https://doi.org/10.1007/s10681-021-02866-0.

Yoon JB, Yang DC, Do JW, Park HG. 2006. Overcoming two post-fertilization genetic barriers in interspecific hybridization between *Capsicum annuum* and *C. baccatum* for introgression of anthracnose resistance. Breed Sci. 56(1):31–38. https://doi.org/10.1270/jsbbs.56.31.